

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	25 July 2026
Name	Shaik Suhail
Project Name	Automated Network Request Management in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Network Service Request Submission	• Request submission through Service Portal • Dynamic UI Policies to show/hide fields (IPs, Ports, Device) • Form validation via IPv4 Regex and mandatory checks
FR-2	Approval Workflow Governance	• Approval routing via Manager for standard access • Approval routing via Network Security Group for administrative access • Approval confirmation via dynamic Email Action buttons
FR-3	Task Orchestration & Fulfillment	• Automated generation and assignment of sc_task to Network Ops • Variable mapping into engineer fulfillment views

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		• Real-time state synchronization between sc_req_item and sc_task
FR-4	CMDB Integration & SLA Tracking	• Target device CMDB mapping (cmdb_ci_netgear) upon completion • Automated SLA tracking (8h High/Critical, 24h Standard) • Executive reporting via Performance Analytics dashboards

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The Service Portal catalog item form must be intuitive, requiring under 3 clicks to submit a complete network request with real-time field validation messages.
NFR-2	Security	Role-based Access Controls (RBAC) and ACLs must restrict request visibility to the opening user, assigned approvers, and authorized Network Engineers.
NFR-3	Reliability	Flow Designer workflows must guarantee zero dropped requests with audit-ready log generation for all approval actions (sysapproval_approver).

NFR No.	Non-Functional Requirement	Description
NFR-4	Performance	Catalog item page load time on Service Portal must remain under 2 seconds, and Flow Designer actions must execute within 3 seconds of submission.
NFR-5	Availability	The ServiceNow Service Catalog interface and background workflows must maintain 99.9% uptime during standard business operating hours.
NFR-6	Scalability	The solution architecture must comfortably support up to 500 concurrent request submissions per hour without impacting instance processing performance.